KAIZEN MATHS

Lesson starter

Trigonometric Equation Solver

Linear Trig Equations

Name _____

S1

Solve $\cos(2x) = -1$ for $0 \leq x < 2\pi$.

S2

Solve $4\sin(2x) = 4\frac{1}{2}$ for $0 \leq x < 2\pi$.

S3

Solve $\sin(3x + \frac{\pi}{3}) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

S4

Solve $\cos(3x) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

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Exit ticket

Trigonometric Equation Solver

Linear Trig Equations

Name _____

E1

Solve $2\cos(3x) = 21$ for $0 \leq x < 2\pi$.

E2

Solve $\cos(3x + \frac{\pi}{4}) = 1$ for $0 \leq x < 2\pi$.

E3

Solve $\sin(2x) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

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Trigonometric Equation Solver

Linear Trig Equations

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Solve $\cos(3x + \frac{\pi}{4}) = 1$ for $0 \leq x < 2\pi$.

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Solve $\sin(2x) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

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Teacher answers & working

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S1

Solve $\cos(2x) = -1$ for $0 \leq x < 2\pi$.

Answer:

$x = \frac{\pi}{2}, \frac{3\pi}{2}$.

Working:

Let $\theta = 2x$.

Solve $\cos(\theta) = -1$ to get $\theta = \pi$.

Now solve $2x = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = \frac{\pi}{2}, \frac{3\pi}{2}$.

S2

Solve $4\ \sin(2x) = 4\frac{1}{2}$ for $0 \leq x < 2\pi$.

Answer:

$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$.

Working:

Divide first, then Let $\theta = 2x$.

Solve $\sin(\theta) = \frac{1}{2}$ to get $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$.

Now solve $2x = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$.

S3

Solve $\sin(3x + \frac{\pi}{3}) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

Answer:

$x = 0.96, 1.48, 3.05, 3.58, 5.15, 5.67$.

Working:

Let $\theta = 3x + \frac{\pi}{3}$.

Solve $\sin(\theta) = -\frac{\sqrt{2}}{2}$ to get $\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$.

Now solve $3x + \frac{\pi}{3} = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = 0.96, 1.48, 3.05, 3.58, 5.15, 5.67$.

Teacher answers & working

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S4

Solve $\cos(3x) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

Answer:
$x = \frac{\pi}{4}, \frac{5\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}, \frac{7\pi}{4}$.

Working:

Let $\theta = 3x$.

Solve $\cos(\theta) = -\frac{\sqrt{2}}{2}$ to get $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$.

Now solve $3x = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = \frac{\pi}{4}, \frac{5\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}, \frac{7\pi}{4}$.

E1

Solve $2\cos(3x) = 21$ for $0 \leq x < 2\pi$.

Answer:
$x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$.

Working:

Divide first, then Let $\theta = 3x$.

Solve $\cos(\theta) = 1$ to get $\theta = 0$.

Now solve $3x = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$.

E2

Solve $\cos(3x + \frac{\pi}{4}) = 1$ for $0 \leq x < 2\pi$.

Answer:
$x = \frac{7\pi}{12}, \frac{5\pi}{4}, \frac{23\pi}{12}$.

Working:

Let $\theta = 3x + \frac{\pi}{4}$.

Solve $\cos(\theta) = 1$ to get $\theta = 0$.

Now solve $3x + \frac{\pi}{4} = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = \frac{7\pi}{12}, \frac{5\pi}{4}, \frac{23\pi}{12}$.

Teacher answers & working

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E3

Solve $\sin(2x) = -\frac{\sqrt{2}}{2}$ for $0 \leq x < 2\pi$.

Answer:

$x = 1.96, 2.75, 5.11, 5.89.$

Working:

Let $\theta = 2x$.

Solve $\sin(\theta) = -\frac{\sqrt{2}}{2}$ to get $\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$.

Now solve $2x = \theta$ for each angle and keep $0 \leq x < 2\pi$: $x = 1.96, 2.75, 5.11, 5.89.$